VeriStrata CEIE — Forensic Investigation Dossier

Case ID: CASE-TAMPERED-A9EE9D99 | **Bundle Type:** LOAN_APPLICATION | **Timestamp:** 2026-11-09T19:57:10 UTC
Forensic Framework: ISO/IEC 27037 & NIST SP 800-86 Cryptographic Evidence Standard

7.1 Document Overview & SHA-256 Source Manifest

Document Source	Classification	Pages	SHA-256 Digest
01_application_form.pdf	APPLICATION_FORM	1	1637175c5682b201b26374...
02_salary_slip.pdf	SALARY_SLIP	1	5bd44044017069def3df14...
03_bank_statement.pdf	BANK_STATEMENT	1	b731d10cfa38d4b1614499...
04_employer_declaration.p	EMPLOYER_DECLARATION	1	ab4c760cdb296923608fcd...

7.2 Extracted Information & Spatial Anchors

Field	Source Doc	Extracted Value	Confidence
account_holder	03_bank_statement.pd	account_holder: Rohan Me	95%
applicant_name	01_application_form.	applicant_name: Rohan Me	95%
application_date	01_application_form.	application_date: 2026-0	95%
declared_income	01_application_form.	declared_income: INR 95,	95%
deductions	02_salary_slip.pdf	deductions: INR 13,000	95%
employee_name	02_salary_slip.pdf	employee_name: Rohan Meh	95%
employee_name	04_employer_declarat	employee_name: Rohan Meh	95%
employer_name	01_application_form.	employer_name: Northstar	95%

7.3 Contextual Analysis & SMT Theorem Prover Reasoning

• **Hypothesis [split_credit]:** **REJECTED** (Solver: z3-5.3.3) — Mathematical Contradiction Isolated: Bank payroll credits total INR 65,000.00, leaving an irreconcilable residual gap of INR 30,000.00 (UNSAT).
• **Hypothesis [posting_delay]:** **REJECTED** (Solver: z3-5.3.3) — Extended Posting Window Ineffective: Scanning 14-day expanded clearing window yields zero additional employer credits; residual gap persists.
• **Hypothesis [gross_net_confusion]:** **REJECTED** (Solver: z3-5.3.3) — Gross Pay Mismatch: Payslip gross pay (INR 78,000.00) also fails to reconcile declared compensation (INR 65,000.00).

7.4 Visual & Structural Multi-Strata Forensic Findings

• **[FILE_STRUCTURE] pdf_incremental_revision** on *01_application_form.pdf* [Severity: **MEDIUM**, Score: 2.000]
• **[VISUAL_PIXEL] ela** on *01_application_form.pdf* [Severity: **MEDIUM**, Score: 0.204]
• **[FILE_STRUCTURE] pdf_incremental_revision** on *02_salary_slip.pdf* [Severity: **MEDIUM**, Score: 2.000]
• **[VISUAL_PIXEL] ela** on *02_salary_slip.pdf* [Severity: **MEDIUM**, Score: 0.227]
• **[FILE_STRUCTURE] pdf_incremental_revision** on *03_bank_statement.pdf* [Severity: **MEDIUM**, Score: 2.000]

7.5 Evidence, Minimal Witness Sets & Traceability

Finding: salary_claim_conflict [Status: **POTENTIAL_MANIPULATION** | Priority: high_review_priority]
Minimal Contradiction Witness Claims:

- net_pay: **net_pay: INR 95,000** (Source: 02_salary_slip.pdf, BBox: [0.4118, 0.2416, 0.4861, 0.2572]) [Crop: /api/claims/bd39a414-bfab-46f5-8a9c-41ff1b059ebd/crop]
- monthly_net_pay: **monthly_net_pay: INR 65,000** (Source: 04_employer_declaration.pdf, BBox: [0.4118, 0.2188, 0.4861, 0.2345]) [Crop: /api/claims/cb13ea55-feee-4c68-b2ec-e22040d108e7/crop]
- salary_credit_row: **04-sep-2026 northstar sal part2 25,000.00 cr** (Source: 03_bank_statement.pdf, BBox: [0.4118, 0.2188, 0.7706, 0.2345]) [Crop: /api/claims/1149be17-abbc-468a-93f9-befeacde741f/crop]
- salary_credit_row: **02-sep-2026 northstar sal 40,000.00 cr** (Source: 03_bank_statement.pdf, BBox: [0.4118, 0.1961, 0.7191, 0.2117]) [Crop: /api/claims/07eca83f-9c48-4c1d-8cfd-60e36ad0cb6b/crop]
Reviewer Adjudication: **ACCEPT** by auditor_automated (Reason: Verified against banking credits and multi-strata forensic telemetry.) — Cryptographically Chained

7.6 Investigation Summary, Standards & Limitations

Overall Synthesis: POTENTIAL_MANIPULATION: Unresolved mathematical SMT conflict with corroborated multi-strata forensic mutations.

Applicable Standards: ISO/IEC 27037: Guidelines for identification, collection, acquisition, and preservation of digital evidence. • ISO/IEC 27041 / 27042: Guidance on assuring suitability and interpretation of digital investigative methods. • NIST SP 800-86: Guide to Integrating Forensic Techniques into Incident Response. • OWASP Machine Learning Security Top 10: Robustness, data integrity, and explainability considerations.

System Limitations: OCR quality may affect extraction confidence on low-contrast, skewed, or degraded scans; raw bounding boxes are preserved. • Visual pixel signals reflect recompression and quantization variance; contextual verification is strictly required. • Z3 SMT solver evaluates domain-valid arithmetic and relational hypotheses; UNKNOWN outcomes are escalated to human review. • Human reviewer oversight is preserved for all consequential legal and credit underwriting decisions.

Root Dossier Integrity Hash (SHA-256): `db99c61f71cf34799cbe2483c8772a58f2192f46ce32bebd58987419eb3701fc`